7 **(a)** energy required to (completely) separate the nucleons (in a nucleus) .... B1 [1]

(b) **(i)** U labelled near right-hand end of line B1
Ba and Kr in approximately correct positions .... B1 [2]

(ii) binding energy is $A \times E_B$ B1
either binding energy of U < binding energy of (Ba + Kr)
or E_B of U < E_B of (Ba + Kr) .... B1 [2]

(c) Krypton-92 reduced to 1/8 in 9 s M1
in 9 s, very little decay of Barium-141 .... M1
so, approximately 9 s .... A1 [3]
OR
$\lambda_{Kr} = 0.231$ or $\lambda_{Ba} = 6.42 \times 10^{-4}$ (M1)
$8 = e^{-\lambda B \times t}/e^{-\lambda K \times t}$ (C1)
$t = 9.0$ s (A1)

Section B

8 (a) (i) 9 V